

The background of the slide is a dense field of red play button icons, which are the standard YouTube logo. The icons are of varying sizes and are scattered across the entire background, creating a textured, repetitive pattern. The overall color scheme is a deep red.

# What factors drive **YouTube** views, and how can Youtuber act on them?

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——Empowering YouTubers with insights to make smarter content choices.



 Youtuber

 Challenges

 Issues

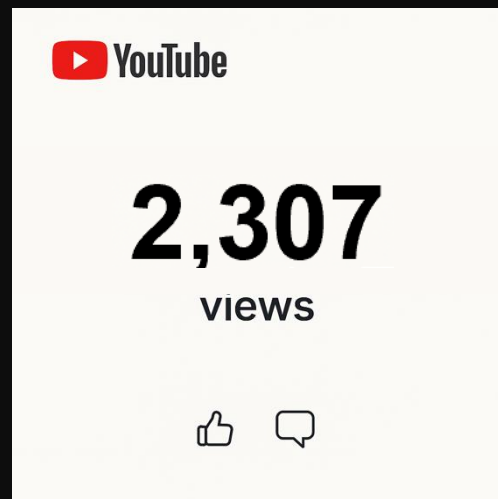
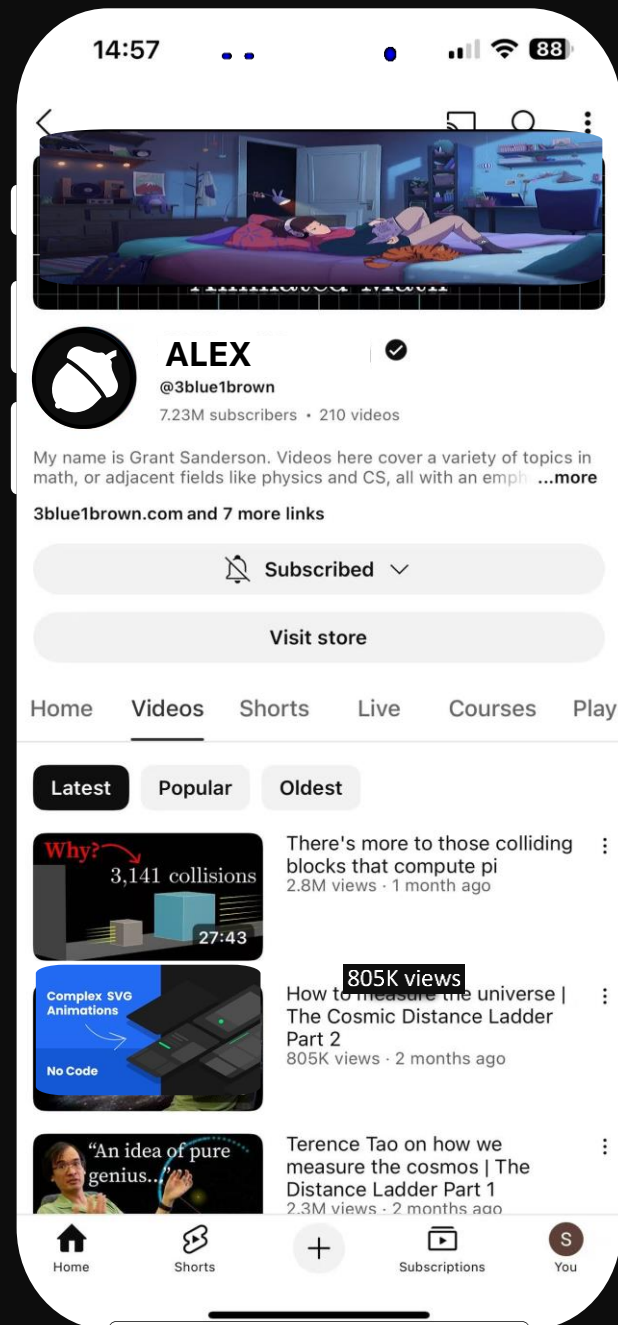
 Project Inspiration

 System Processing

 Application

 Challenges

 Q&A



Alex, a full-time YouTube creator.

Like many creators out there,  
every day, Alex gets stressed about  
whether his videos will get enough  
views

# Challenges

Every day, he spends hours brainstorming content ideas, designing thumbnails, and choosing when to publish. But he still has many problem:



What if I miss peak hours?



Does weekend posting help?



Is timing strategy different across regions?



o certain categories perform better?



Is text on thumbnails helpful?



How much does thumbnail alignment affect CTR?



Do faces increase attention?

## Time

When is the best time to publish to maximize visible?

Should I post in the morning or at night?

## Content

Which topics drive the most engagement??

How does title wording affect clicks?

## Thumbnail

Are bright colors better than minimalism?

Should I test different styles before posting?

And once a video is published, it's nearly impossible to tweak key elements like title, thumbnail, or timing without risking performance loss.

# Issue: Uncertain performance after publishing

Sometimes his content unexpectedly flops. Other times, videos he thought were “meh” go viral. This unpredictability frustrates him-and wastes time and money.

| Video                                                                                                                                                                                  | Visibility | Date ↓       | Views  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|--------|
|  <p><b>GT3 Thrill Ride</b><br/>Experience the roar of a Porsche GT3 RS in full throttle.</p>         | Public     | Aug 02, 2023 | 1100 ? |
|  <p><b>Monkey Mission: Start to Finish</b><br/>Join the monkeys on a quirky action-packed quest.</p> | Public     | Jun 05, 2023 | 60k    |
|  <p><b>Midnight Streamer Life</b><br/>A late-night look into the life of a digital dreamer.</p>     | Public     | Jun 02, 2023 | 200 ?  |
|  <p><b>Build Animated SVGs</b><br/>Create stunning SVG animations—no code needed.</p>              | Public     | Feb 23, 2023 | 230k   |



# Project Inspiration

Alex starts wondering:

"What if I had a tool that could predict video views before publishing based on my thumbnail, title, content and timing settings this kind of stuffs?"

"Could it even suggest adjustments to help me improve performance before it's too late?"



## Predict Views Before Publishing

We aim to build a tool that allows YouTubers to **estimate a video's views in advance**, using both metadata and thumbnails—so creators can test ideas before going live.



## Providing Actionable Feedback on Content Settings

Beyond prediction, our model will **highlight which settings—like time, category, or thumbnail—are hurting or helping performance**, providing targeted suggestions for improvement.

TWO GOALS

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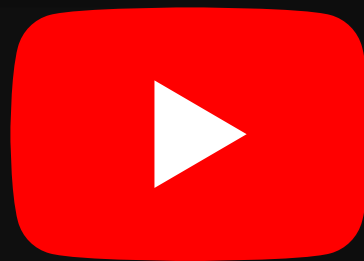
 Project Inspiration

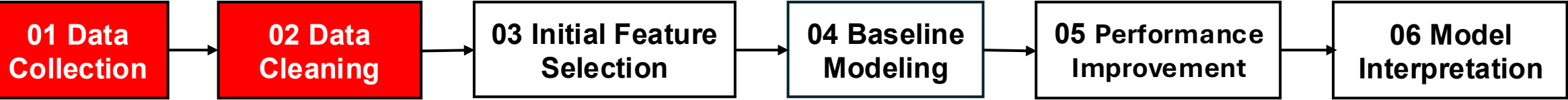
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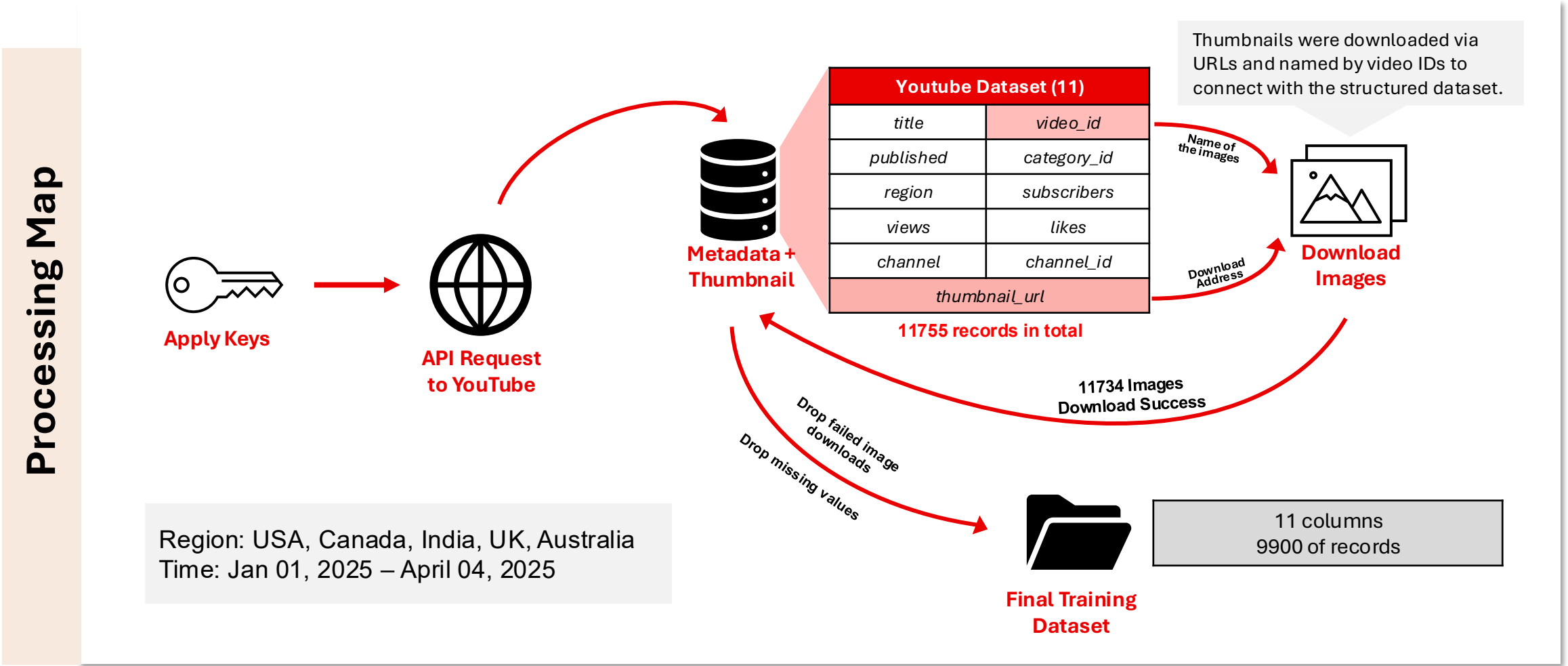
 Challenges

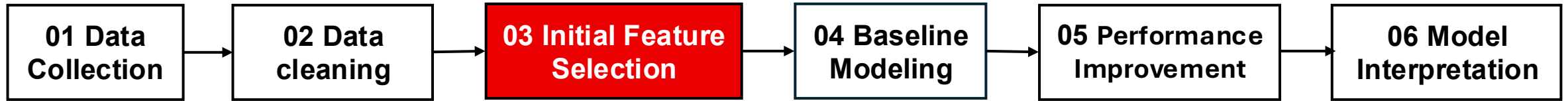
 Q&A



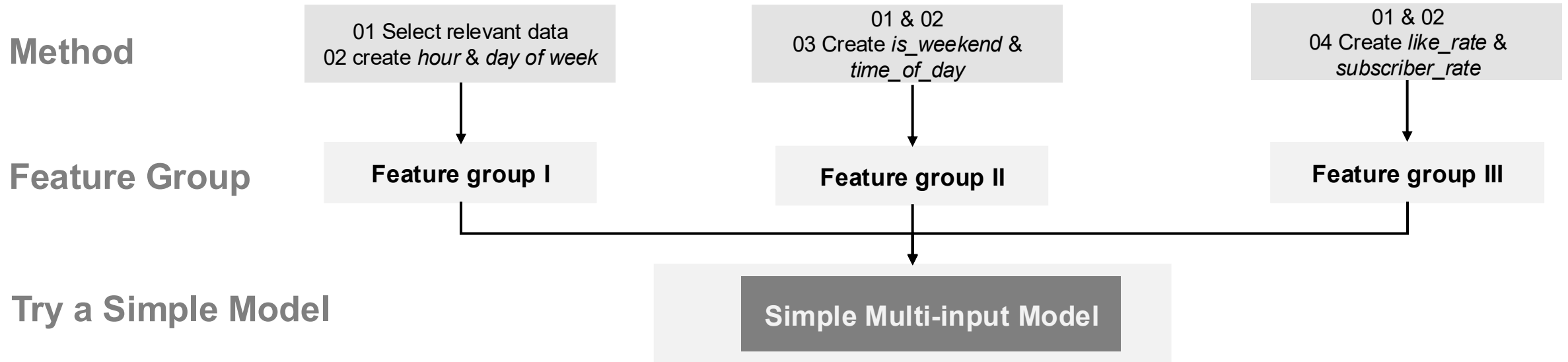


# 01 Data Collection & 02 Data Cleaning



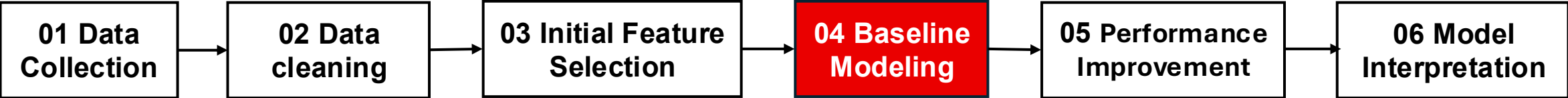


## 03 Initial Feature Selection

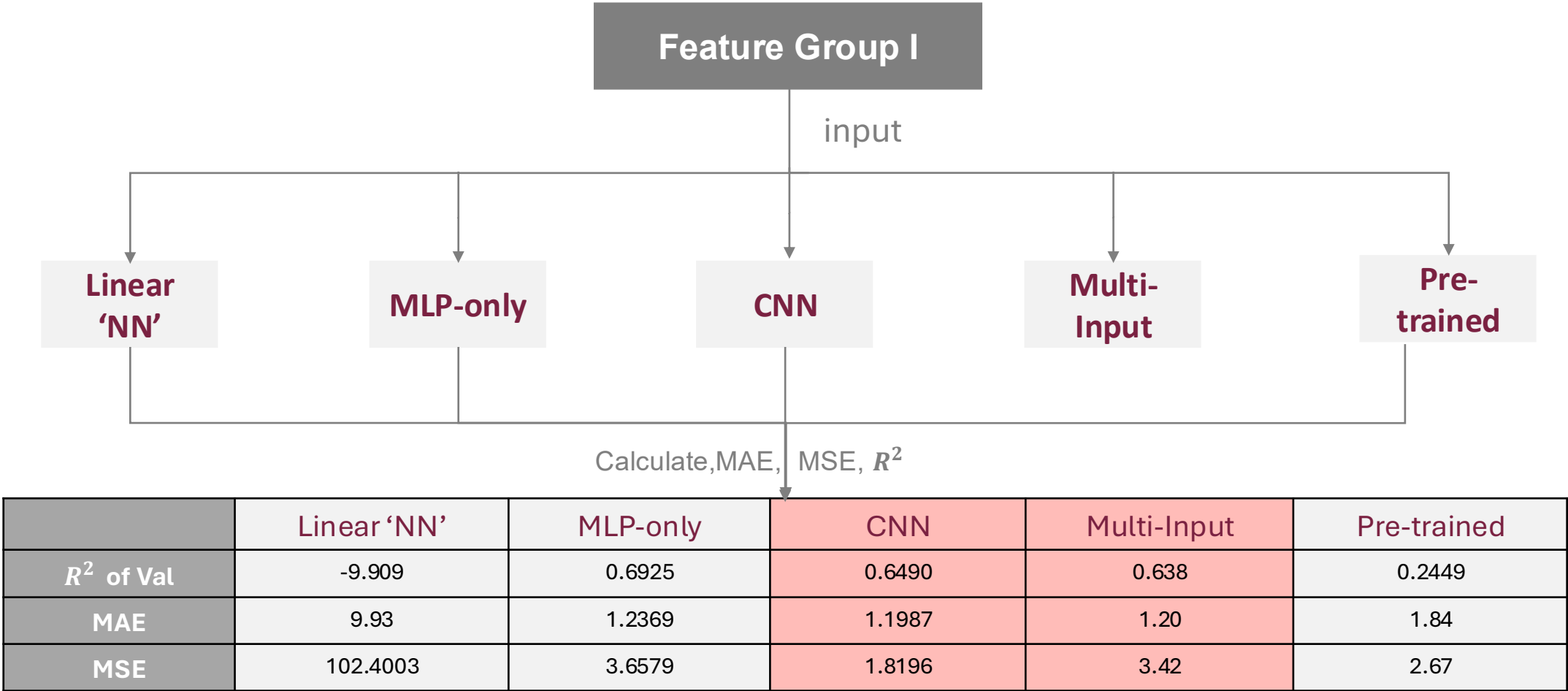


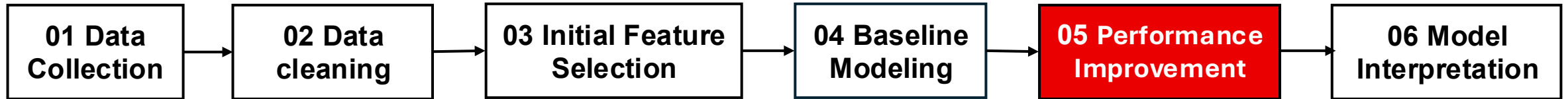
| Feature Group | I (Best Feature group) | II    | III   |
|---------------|------------------------|-------|-------|
| $R^2$         | 0.595                  | 0.583 | 0.538 |
| MSE           | 3.91                   | 3.93  | 4.36  |
| MAE           | 1.07                   | 1.18  | 1.36  |





04 Baseline Modeling





## 05 Performance Improvement – Feature Extraction

### Thumbnail

We used public API to extract features from image data

| New Feature          | Description                         |
|----------------------|-------------------------------------|
| <i>Has Face</i>      | Whether a face is detected (0 or 1) |
| <i>Face Count</i>    | Number of faces detected            |
| <i>Has Text</i>      | Whether text is detected (0 or 1)   |
| <i>Detected Text</i> | Text content detected in the image  |

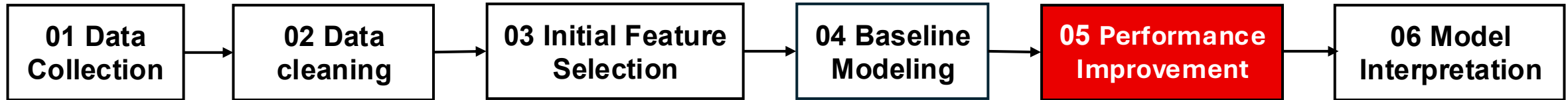
Original  $R^2 = 0.638$

Current  $R^2 = 0.478$

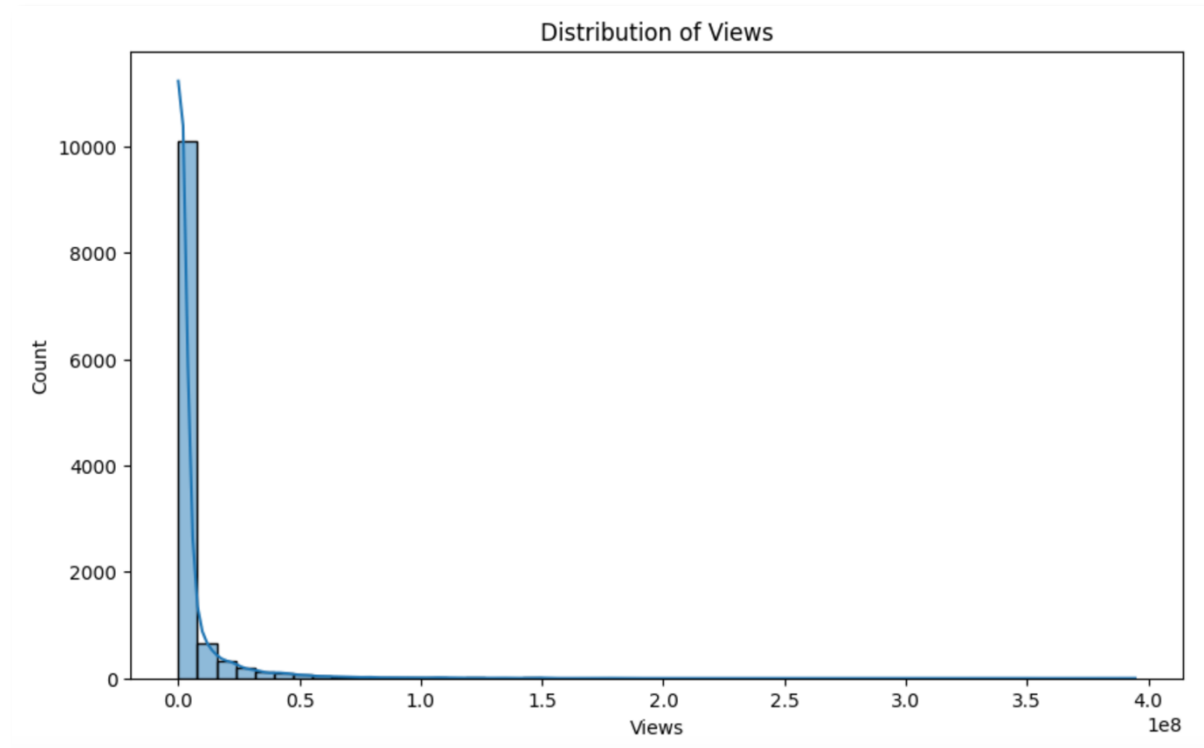
### Headline

We try to detect features based on text data

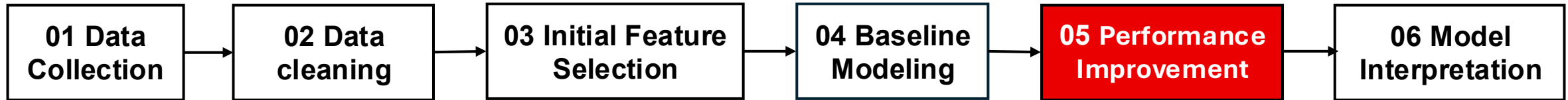
| New Feature                                                  | $R^2$        |
|--------------------------------------------------------------|--------------|
| No Title Feature                                             | 0.638        |
| + Symbol and Special Character Count                         | 0.626        |
| <b>+ Symbol Character<br/>(with Original Tabular Branch)</b> | <b>0.656</b> |
| + Symbol Character<br>(with Small Tabular Branch Upsize)     | 0.621        |



## 05 Performance Improvement – Target Transformation

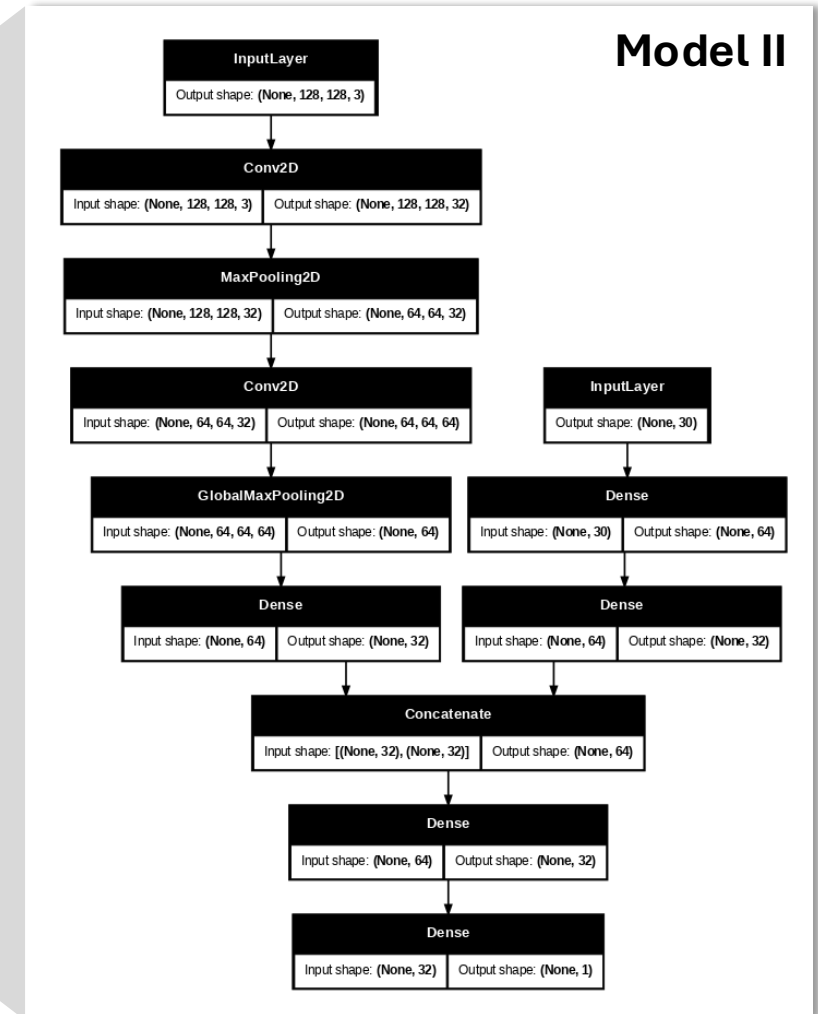


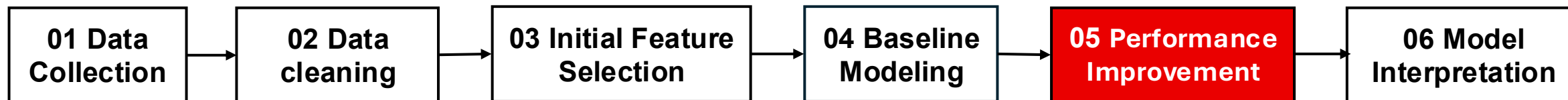
|                 | $R^2$ of train | $R^2$ of validation | $R^2$ of train (raw view) | $R^2$ of validation (raw view) |
|-----------------|----------------|---------------------|---------------------------|--------------------------------|
| $y' = \log(y)$  | 0.783          | 0.656               | huge negative             | huge negative                  |
| $y' = \sqrt{y}$ | 0.808          | 0.794               | 0.715                     | 0.763                          |



## 05 Performance Improvement – Model Designing

|                           | Initial Model                                           | Model II                                                                              | Model III                                                                                |
|---------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Image Branch</b>       | Flatten → Dense(32)                                     | Conv2D(32) → MaxPool → Conv2D(64) → GlobalMaxPooling → Dense(32)                      | Conv2D(32) → MaxPool → Conv2D(64) → MaxPool → Conv2D(128) → GlobalMaxPooling → Dense(64) |
| <b>Tabular Branch</b>     | Dense(64) → Dense(32)                                   | Dense(64) → Dense(32)                                                                 | Dense(64) → BatchNormalization → Dropout → Dense(32) → BatchNormalization → Dropout      |
| <b>Fusion and Output</b>  | Concatenate → Dense(32)                                 | Concatenate → Dense(32)                                                               | Concatenate → Dense(64) → Dropout → Dense(32)                                            |
| <b>Key Design Changes</b> | Directly flatten image; simple stacking of Dense layers | Simplified structure; fewer convolution and dense layers to better fit small datasets | Deep convolutional feature extraction; added normalization and dropout for stability     |
| $R^2$ of train            | 0.808                                                   | <b>0.824</b>                                                                          | 0.796                                                                                    |
| $R^2$ of val              | 0.794                                                   | <b>0.809</b>                                                                          | 0.783                                                                                    |
| $R^2$ of train (raw view) | 0.715                                                   | <b>0.729</b>                                                                          | 0.706                                                                                    |
| $R^2$ of val (raw view)   | 0.763                                                   | <b>0.775</b>                                                                          | 0.757                                                                                    |





## 05 Performance Improvement – Hyperparameter Tuning

| Hyperband (Halving)                          |          | Random Search       |               | Bayesian Search |
|----------------------------------------------|----------|---------------------|---------------|-----------------|
| <div><div></div><div></div><div></div></div> |          |                     |               |                 |
|                                              | Original | Hyperband (Halving) | Random Search | Bayesian Search |
| $R^2$ of train                               | 0.824    | 0.821               | 0.815         | <b>0.828</b>    |
| $R^2$ of val                                 | 0.809    | 0.802               | 0.798         | <b>0.811</b>    |
| $R^2$ of train (raw view)                    | 0.729    | 0.734               | 0.771         | <b>0.768</b>    |
| $R^2$ of val (raw view)                      | 0.775    | 0.761               | 0.774         | <b>0.769</b>    |
| Run Time                                     | /        | Aggressive          | Moderate      | Fast            |

### 1. Best Trade-off Between Fit and Generalization

- Achieved a relatively high training  $R^2$  (0.768) and validation  $R^2$  (0.769)
- Indicates stronger learning without significant overfitting.

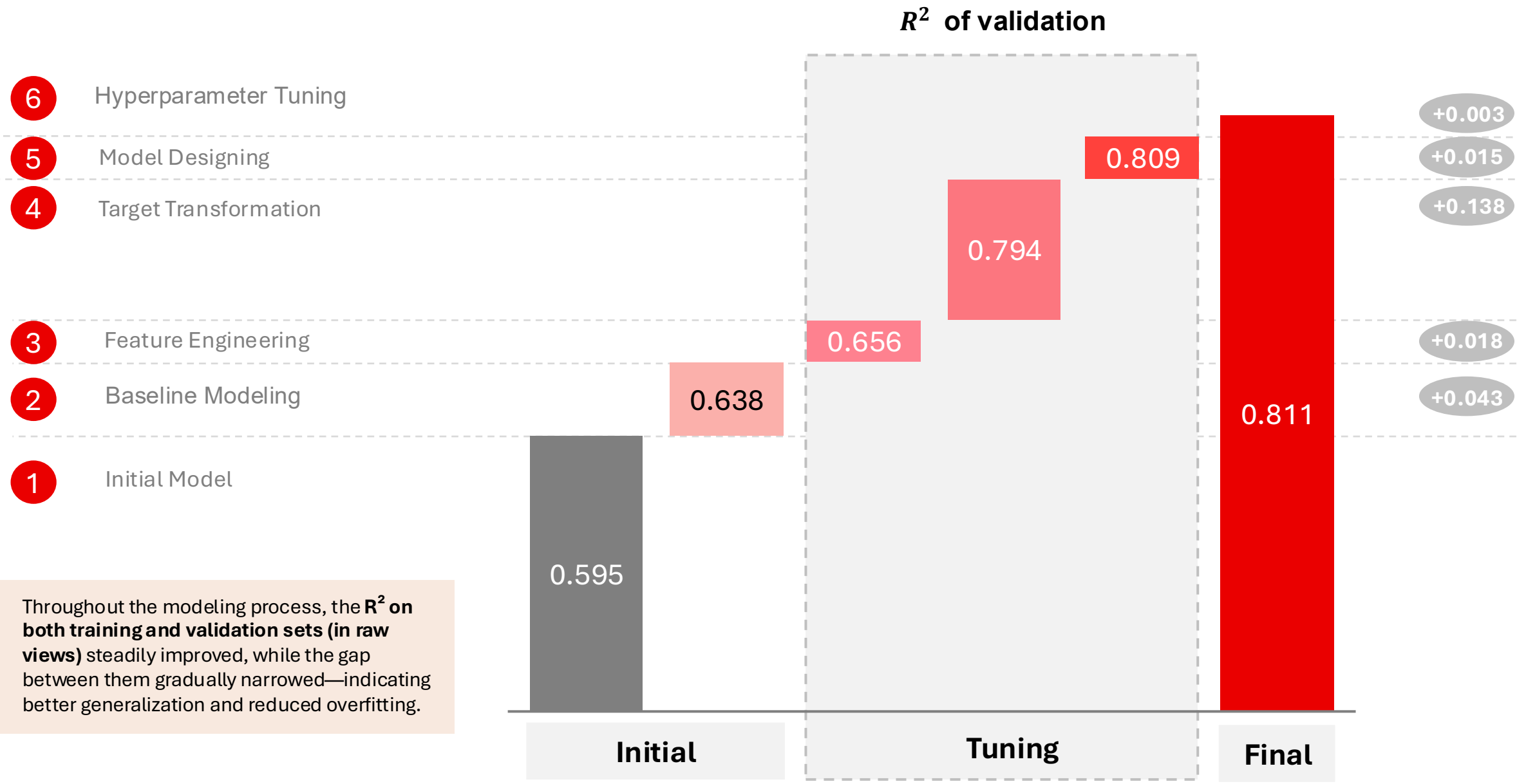
### 2. Industry-Preferred Optimization Strategy

- Bayesian Optimization systematically balances exploration and exploitation.
- Commonly used in state-of-the-art hyperparameter tuning workflows.

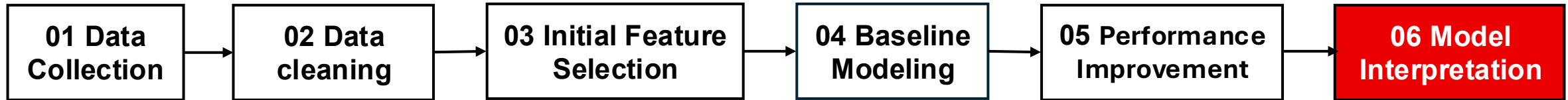
### 3. Efficient Gains with Reasonable Runtime

- While the performance gain is modest, it is consistent across metrics.
- Selected method based on both performance and resource efficiency.

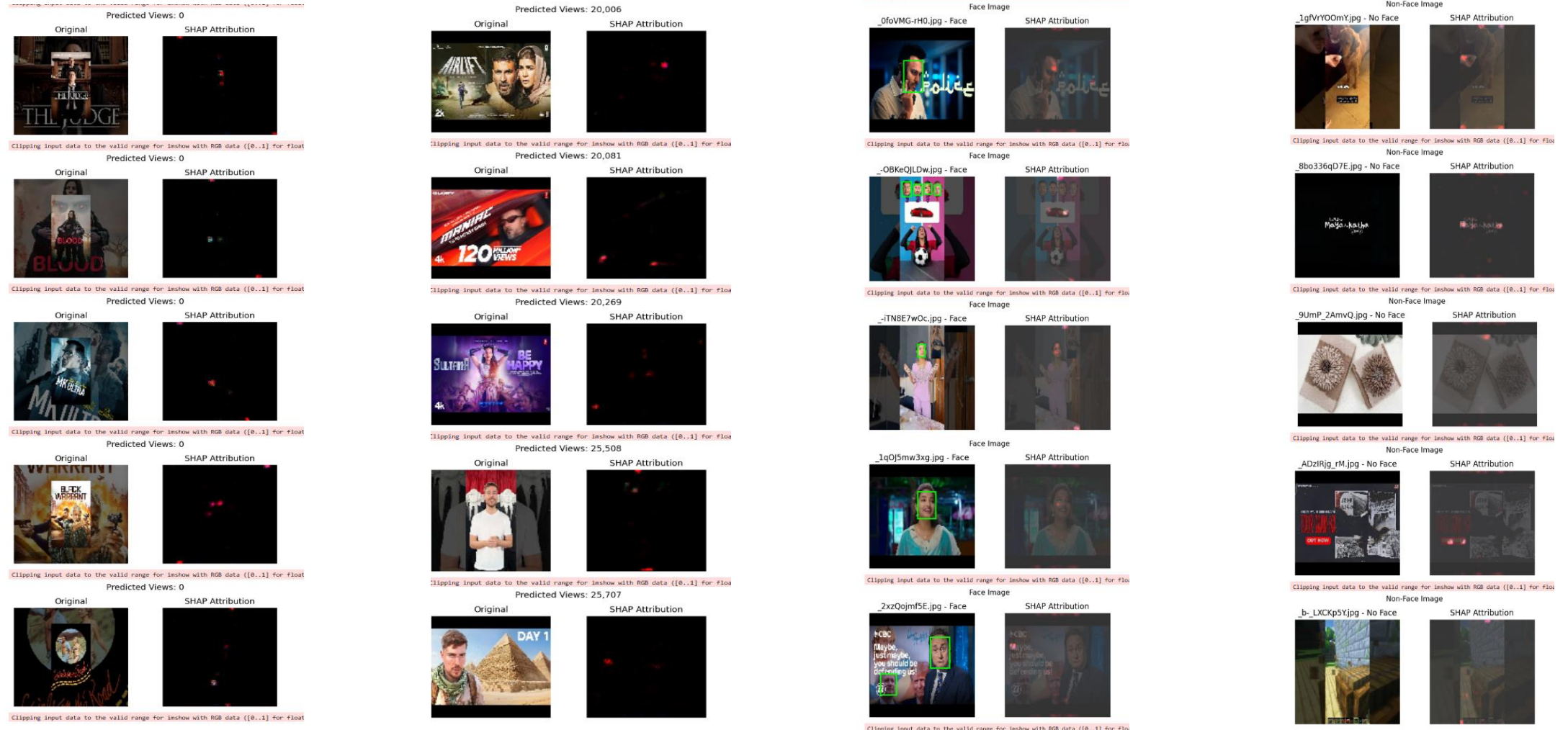
# Step-by-Step Model Improvement Journey

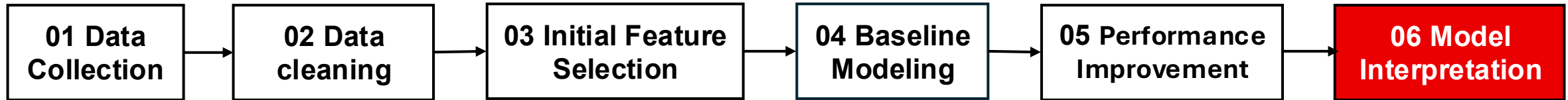






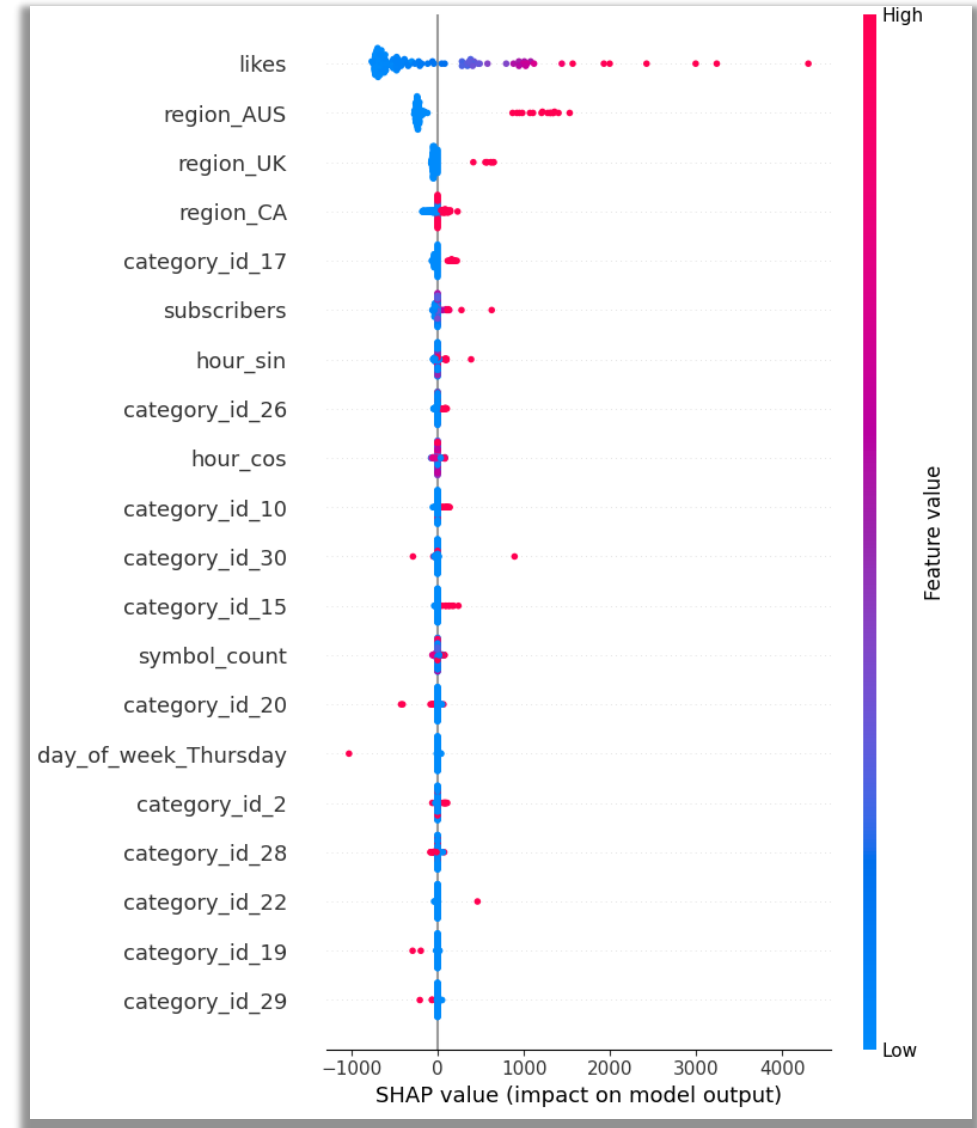
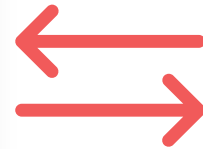
## 06 Model Interpretation -- SHAP





## 06 Model Interpretation -- SHAP

- The number of **likes** is the clearest indicator of a video's popularity in your model.
- The model finds **geographic region (AUS)** to be a significant factor, likely due to different audience sizes or engagement behaviors.
- **"Sports" category** is positively associated with higher predicted views.



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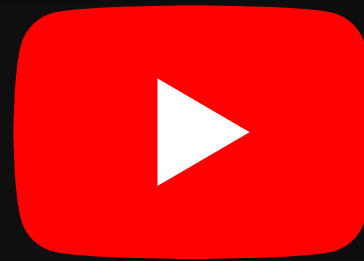
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# Try our system !!!

Tabular Inputs

☐ Do you already know the like count? ?

You can leave Likes blank if predicting before publish; it will default to 0.

Subscribers ?

10000

-

+

Publish Date ?

2025/05/01

Publish Hour (0-23) ?

0

12

23

Region

AUS

Category ?

Autos & Vehicles

Video Title (optional)

Fork

## YouTube Views Predictor

### Upload Thumbnail

Choose a JPG/PNG

Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files

Please upload a thumbnail image.



## Your Face = Your Clicks

Creators should consider including **close-up, expressive facial images** in their thumbnails — especially when the content is personality-driven, emotional, or reaction-based. If not using a face, then use **bold, readable text** or include **visually striking focal elements** (e.g., logos, objects, strong contrast). YouTubers in niches like **education, commentary, or storytelling** should test **face + context** combinations to increase thumbnail effectiveness.

- 1 Optimize content for **emotional or opinion-based reactions**, which encourage likes (e.g., surprises, controversial takes, humor). Use strong **CTAs** in the first 30 seconds (“Like if you agree!”).
- 2 If relevant, **target Australian audiences** with content featuring local slang, time zones, or references.
- 3 Use **seasonal spikes** (e.g., World Cup, Olympics) to capitalize on search trends.

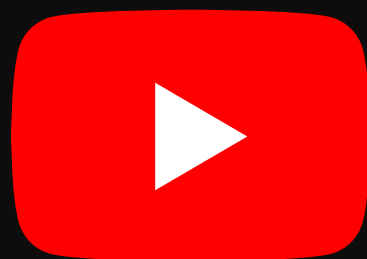
# Challenges

**01 Opaque scraping logic:** Our scraper produced duplicate loops, erratic daily yields, and a pronounced regional skew ( $\approx 5,000$  Canadian records versus only 5,000 combined from the other four regions), making it hard to trust and extend.

**02 API rate constraints:** 10,000 requests per day, making it even harder to experiment with our scraping algorithm.

## 03 Small Datasets and Computational Constraints

**04 Unreliable SHAP explanations:** Attempts to use SHAP for image feature importance returned inconsistent or non-actionable insights, limiting the interpretability of our outputs.



# Thank You!